Elements of Mechanical Engineering

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Energy sources

- Energy sources → sources of energy from which energy can be extracted and utilized for mankind
- Growth of a nation → depends on availability of energy sources
- Classification of energy sources:

(1) Conventional and Non-conventional sources

(2) Renewable and non-renewable sources

Contd….

Conventional energy sources

- These sources have been used traditionally for many years and widely used at present
- Easily converted into mechanical energy
- Likely to be depleted i.e. these have limited availability
- **Examples** → Coal, petrol diesel, nuclear fuels, CNG and LPG

Contd….

Non-conventional energy sources

- These sources are non-traditional, used on a large scale, and not used routinely at present
- Easily can’t be converted into mechanical energy
- Non-depletable i.e. available in vast quantities
- **Examples →** Solar, wind, tidal, geothermal, biogas, etc

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Renewable energy sources

- Energy generated from natural sources and can be generated again and again as and when required
- **Types** → Solar, wind, tidal, geothermal, hydel, etc

Non-renewable energy sources

- Energy generated from sources available on earth → limited quantity → exhaustible sources
- **Types** → Fossil fuel, natural gas, nuclear fuel, oil and coal

Contd….

Advantages of RE over NRE

- RE sources are available in abundant quantity and free to use
- NRE are limited and bound to expire one day
- RE have low carbon emissions
- RE sources are green and environmental friendly

Solar energy

- Solar energy → energy comes from the Sun
- The Sun generates its energy due to continuous fusion reaction of H_2 nuclei into helium taking place in the sun
- A large amount of heat energy in the form of electromagnetic radiation contained in Sun rays
- Heat energy can be utilized to generate electrical power
- Two techniques/systems to convert solar energy into electrical energy → solar thermal systems and solar photovoltaic systems

Solar thermal systems

- Solar thermal systems → provide thermal energy from the Sun rays used for various domestic and industrial fields

- Essential unit of solar thermal system → solar collector

- Solar collector absorbs heat energy from Sun and then transfers to the transport fluid

- Transport fluid delivers heat to storage tank/boiler/heat exchanger to be utilized in the subsequent stages of system

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Classification of solar collectors

- Flat plate collector (Non-concentrating type)
  - absorb solar radiation when it is received on the surface of collector
  - absorbs beam radiation as well as diffused radiation
- Parabolic concentrating collector (Concentrating type)
  - increases the concentration of solar radiation per unit area & then absorbs
  - uses only the beam radiation which converge using optical lens & then absorbs

Contd….

Flat plate collector

Basic elements of a flat plate collector:

- Transparent glass cover (glazing sheet)
- Absorber plate (usually made of Cu, Al or steel)
- Tubes in contact with the absorber plate
- Insulation (made of glass wool) on the bottom and sides to prevent heat losses

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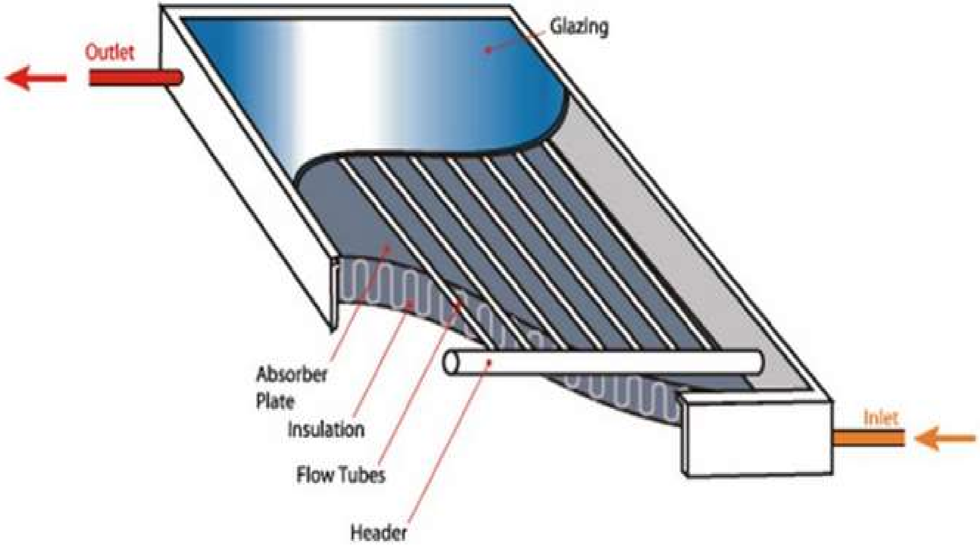
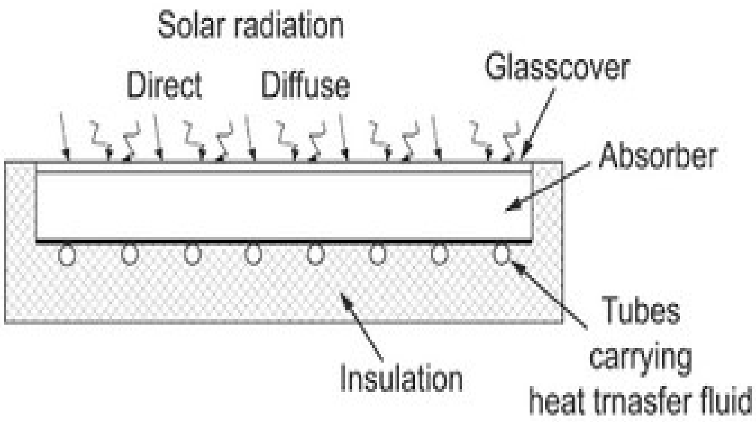
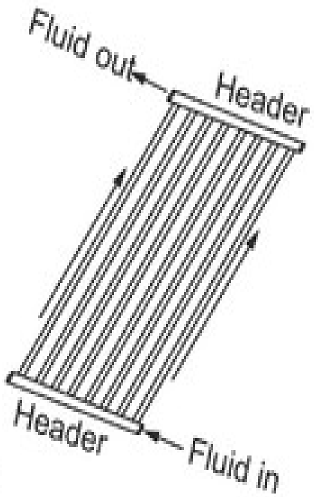
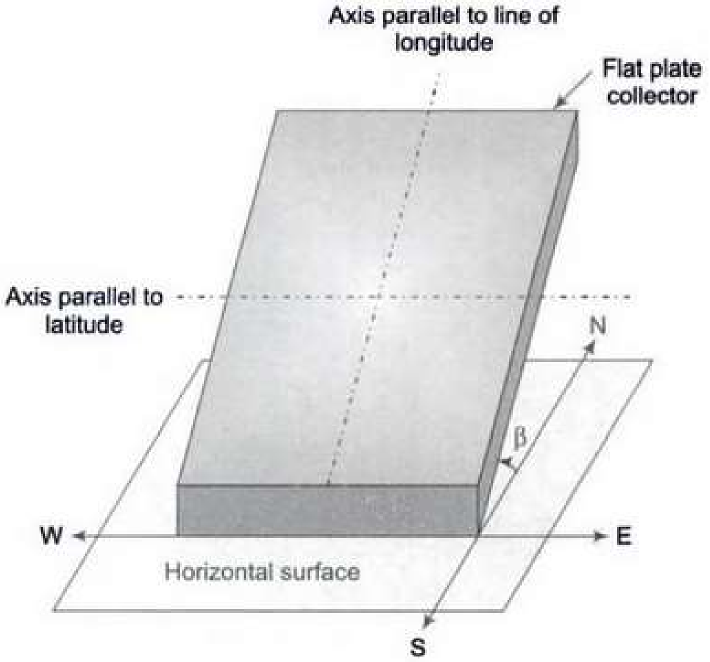

Outlet
Glazing
Absorber Plate
Insulation
Flow Tubes
Header
Inlet


Solar radiation
Direct
Diffuse
Glasscover
Absorber
Tubes carrying heat trnasfer fluid
Insulation


Fluid out
Header
Header
Fluid in


Axis parallel to line of longitude
Flat plate collector
Axis parallel to latitude
N
β
W
E
Horizontal surface
S

Contd….

Advantages of flat plate collector

- Simple in construction and requires less maintenance
- Mechanically strong as it is secured on a rigid platform
- As it is installed outdoor, it can withstand atmospheric disturbances

Disadvantages

- Large amount of heat is lost due to absence of optical system
- High temperatures cannot be attained and results in low heat energy absorbed by collector

Contd….

Evacuated tube collector

- To overcome the limitation of flat plate collectors, evacuated tube collectors are used
- Evacuated tube collector uses transparent glass tubes always perpendicular to sun rays for most of the day
- Allows solar system to operate at much higher efficiency and temperature for a long period of time
- Used particularly in areas with cold and cloudy weathers

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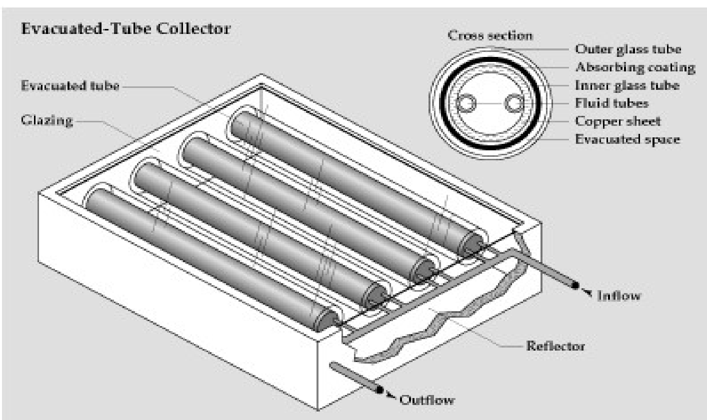
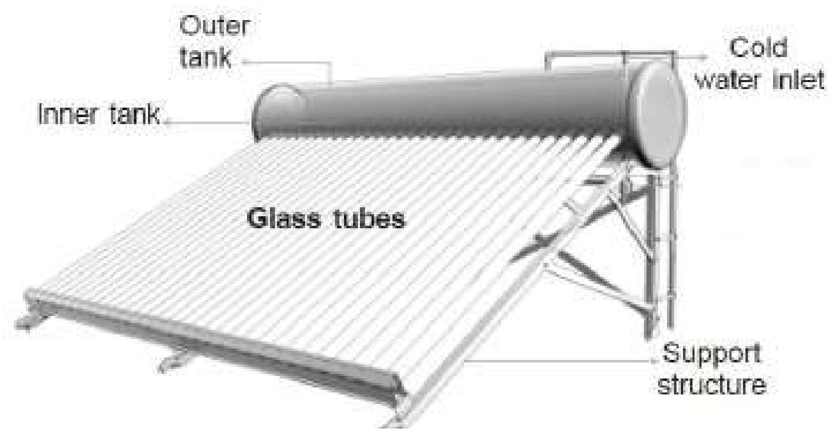
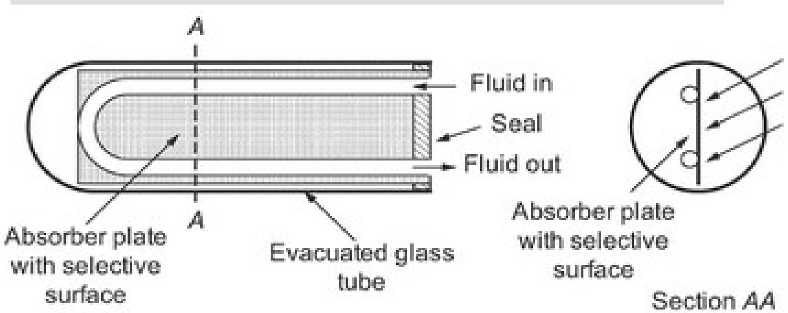
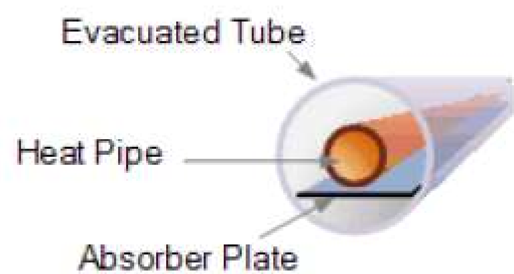
Working principle

- Evacuated tube collector consists of a no. of rows of parallel cylindrical glass tubes supported on a frame
- Each tube consists of two concentric glass tubes made of BS glass









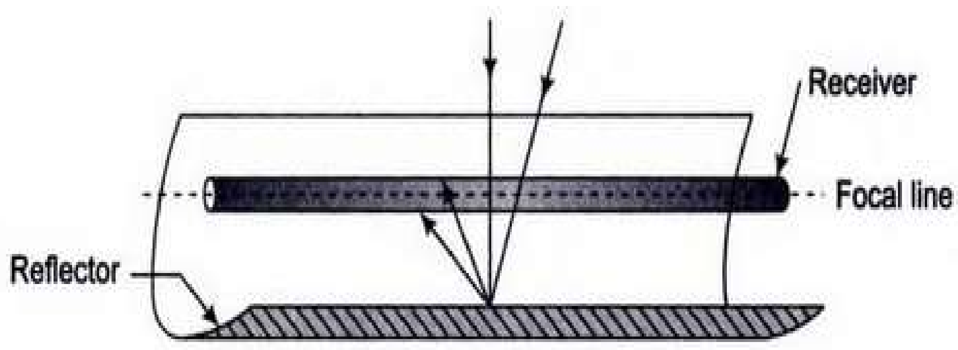
Contd….

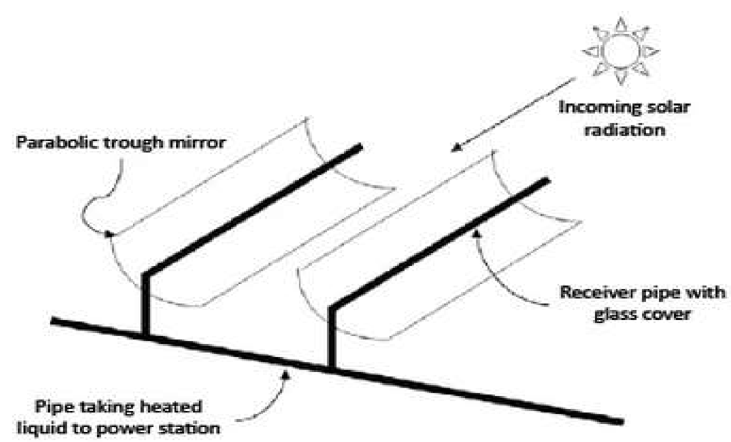
- Outer surface of inner tube coated with selective coating that absorbs excellent solar radiation and inhibits heat loss
- Vacuum is created in between the glass tubes and vacuum acts as an insulator that eliminates conductive and convective heat loss
- Inside each glass tube, a metal absorber plate (Cu/Al) attached with two fluid tubes forming a “U” path for the fluid
- Sun rays passes through outer tube is absorbed by inner tube and the absorber plate is heated up
- Plate transfers heat to the fluid circulating in the fluid tubes, and hot water comes out of the header is transferred to storage tank

Contd….

Parabolic concentrator type collector

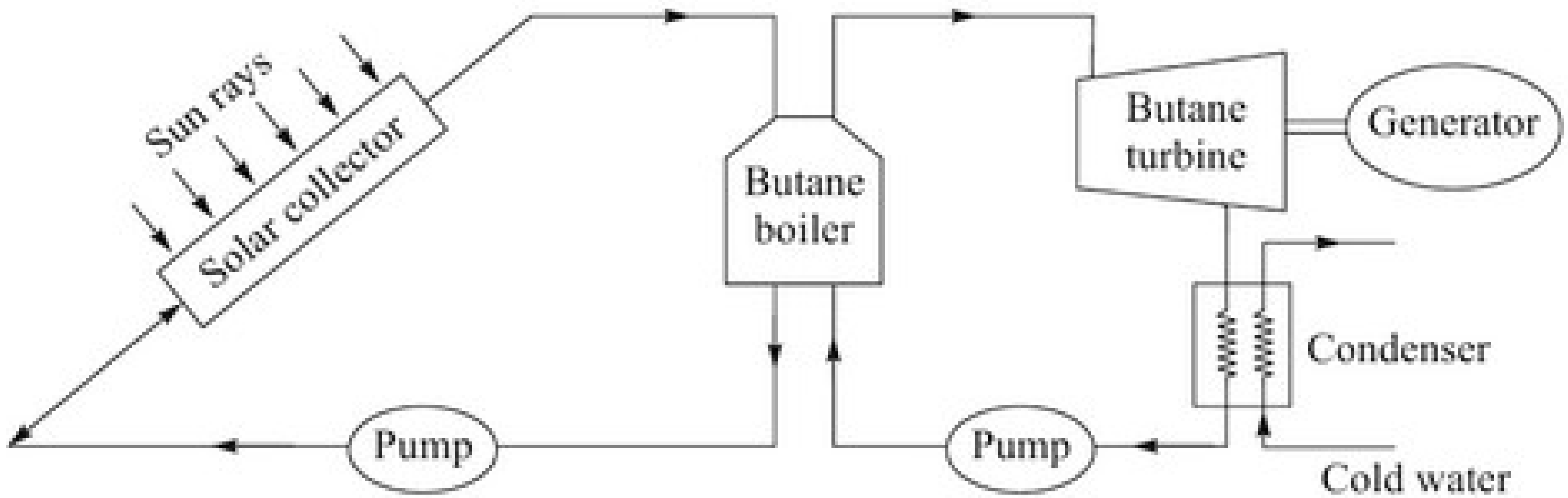
- Consists of a parabolic trough reflector and blackened metal tube receiver at its focal line
- Parabolic reflector is rotated about one axis to track the sun rays
- Solar radiation striking on the reflector surface reflects and the receiver tube absorbs the radiation
- Fluid flowing through receiver is heated and carries to next stage





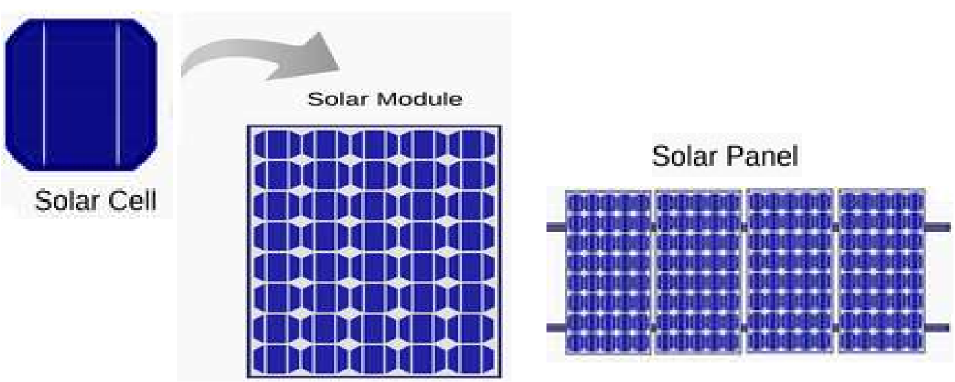
Power generation by solar collector

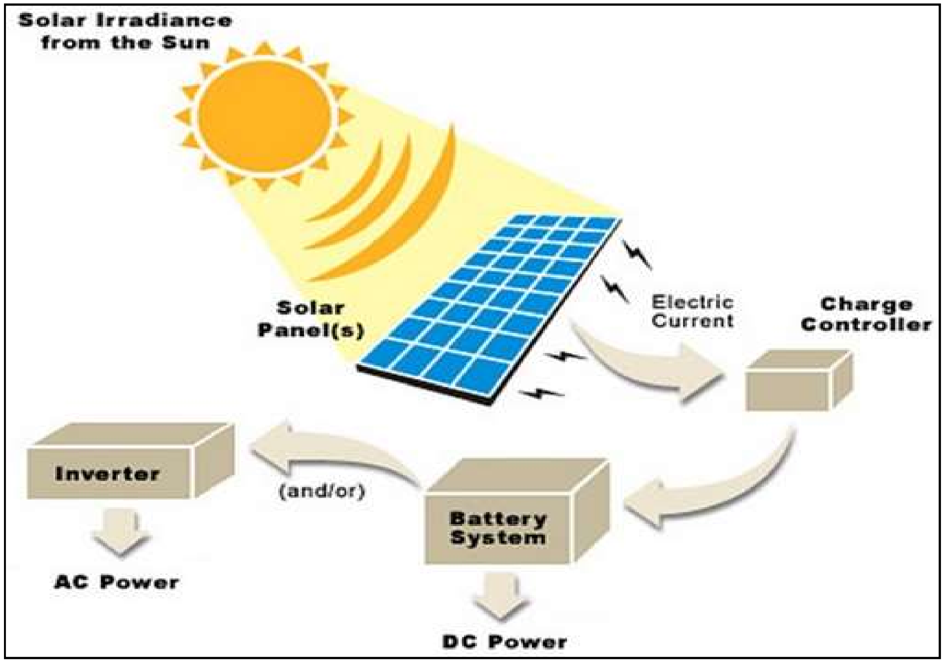
- Sun rays focused on solar collector and heat butane water
- Heated butane water generate butane gas in the boiler
- Butane gas supplied to turbine and mechanical work is obtained
- Generator coupled with turbine generates electrical power



Solar photovoltaic system

- Solar PV systems convert solar energy directly into electrical energy using solar cells
- Solar cells are usually made of silicon → solar cells are intermittent current source and used with batteries to store electricity
- Solar cells are connected in series or parallel to produce required electrical power





Contd….

Advantages of Solar PV

- No thermal-mechanical link and no moving parts
- Reliable and no maintenance

Disadvantages

- Solar cells are of high cost
- Efficiency of solar cells is low
- As solar energy is intermittent, electrical energy storage is required which makes the whole system expensive

Contd….

Applications of Solar PV

- Space satellites
- Remote radio communication
- Booster stations
- Marine warning lights
- Solar powered vehicles

Contd….

Advantages of Solar energy

- Freely available in nature
- Renewable energy source
- Does not pollute the environment

Disadvantages

- Available only during day times and clear days
- Energy can be obtainable or not depends on seasonal variations

Wind energy

- Wind → moving mass of air
- Wind is produced due to uneven heating of atmosphere by sun that creates temperature, density and pressure differences
- Wind energy → K.E. of moving mass of air
- **Wind turbines** → convert wind energy into mechanical energy → mechanical energy to electrical energy with help of generator

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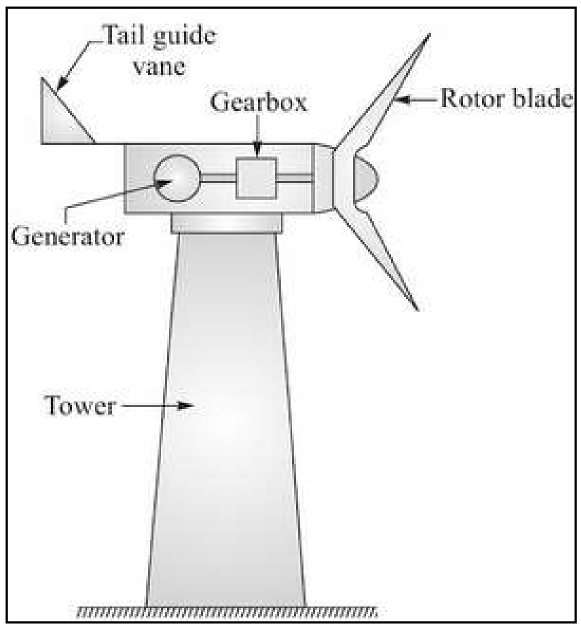
- Classification of wind turbines:

  (1) Horizontal axis wind turbine

  (2) Vertical axis wind turbine

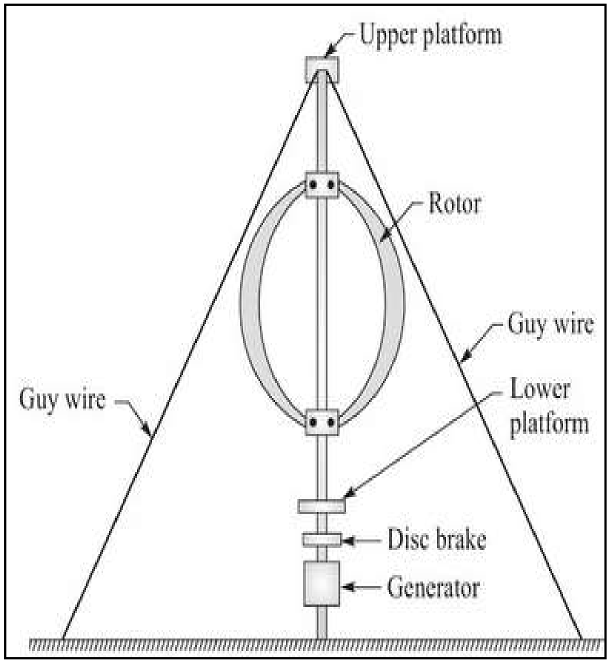
Horizontal axis wind turbine



Axis of rotation of rotor parallel to ground and also wind direction

Contd….

Vertical axis wind turbine



Axis of rotation of rotor perpendicular to ground and also wind direction

Blades of wind turbine → fiber reinforced plastic composite → less cost, ease of manufacturing and high strength to weight ratio

Contd….

Advantages of wind energy

- Freely and abundantly available in nature
- Renewable energy source
- Does not cause pollution to environment
- Less maintenance and operating cost

Disadvantages

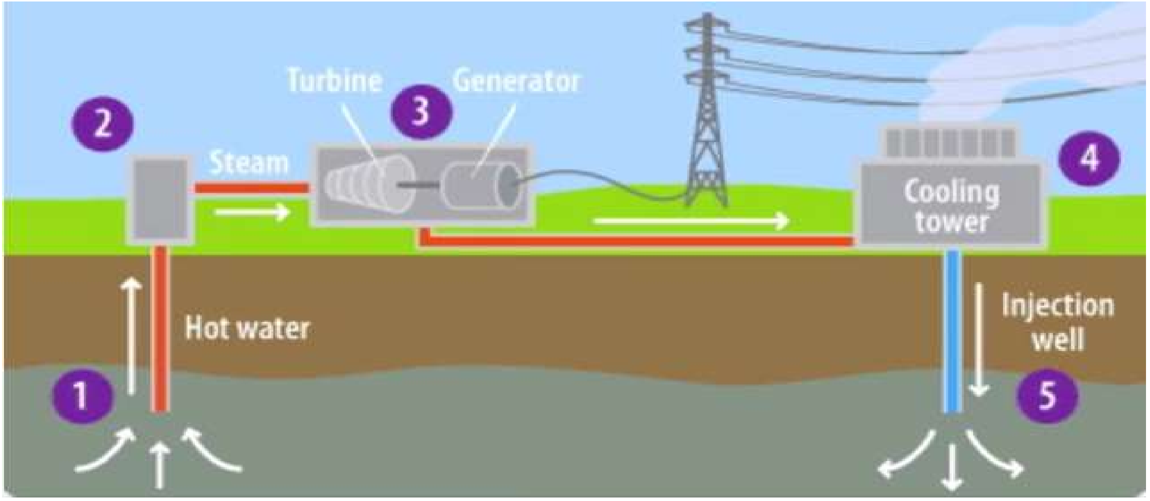
- Cannot produce steady and consistent power
- Generate only low power

Geothermal energy

- Geothermal → heat of the earth
- Geothermal energy → internal heat energy available at a considerable depth below the earth surface
- Heat comes from the molten rock within the earth called magma and temperature is about 3000 °C
- Source of geothermal → magma

Contd….

- At a certain distance below the Earth's surface, water is converted into steam by the heat available at magma
- Steam comes out through the vents of Earth's surface and used to run the turbine
- Turbine runs the generator to produce electric power



Geothermal power plant

Contd….

Advantages

- Energy is continuous available
- More reliable
- Good potential to meet the power requirement
- Low cost

Disadvantages

- Components of the plants liable to be eroded
- Makes pollution environment

Ocean energy

- Ocean energy → energy comes from the ocean
- Energy of ocean can originate either in three forms which can be converted to electrical energy
  - Tides of ocean
  - High K.E. of large waves produced in the ocean by wind
  - Thermal energy of ocean caused due to temperature gradient from ocean surface to the inside depth of ocean

Contd….

Ocean thermal energy conversion (OTEC)

- Water at ocean surface is around 25°C while at a depth of 100 m-200 m is about 5°C
- Temperature gradient between these two levels is 20°C
- Warm water from ocean surface used to heat the low boiling point liquid refrigerant and liquid converted into high pressure vapor

Liquid refrigerant

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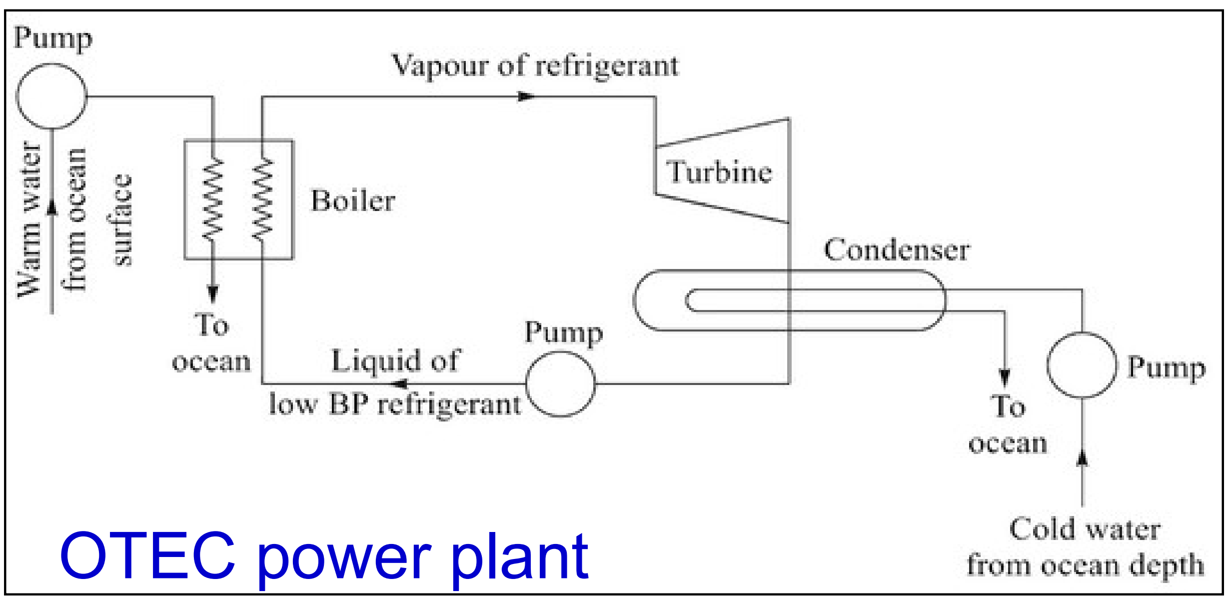
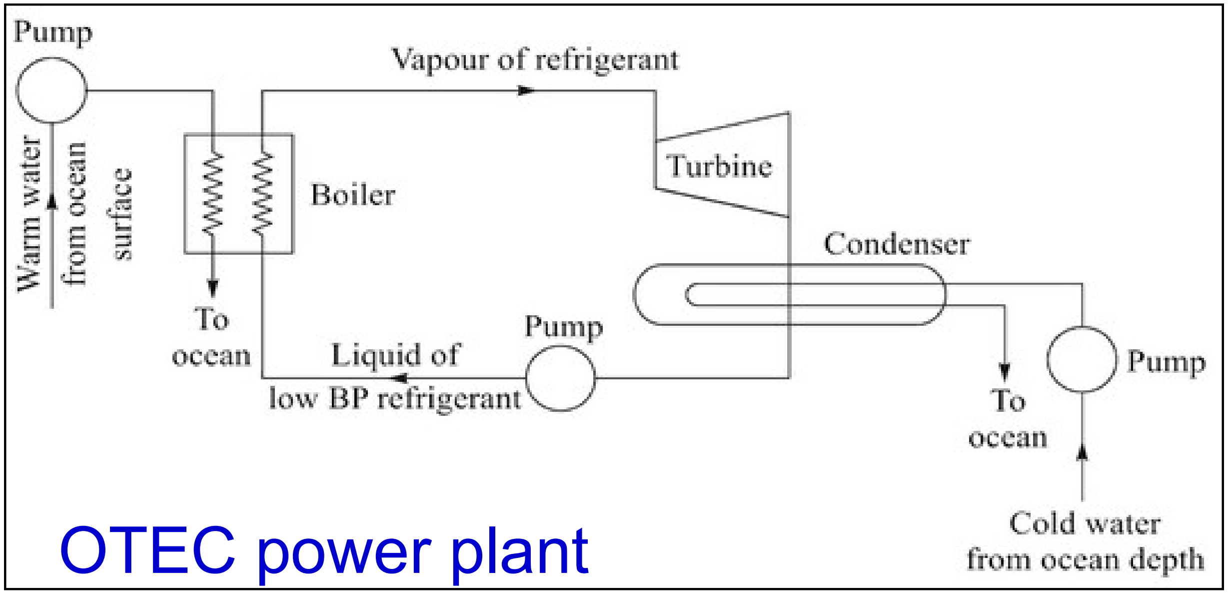
NH_3 and Freon



OTEC power plant

Contd….

- Liquid vapor used to run the turbine and electricity is produced using generator
- Vapor is condensed into liquid in the condenser and pumped back to the boiler



OTEC power plant

Contd….

Advantages of OTEC

- Electrical power generation is continuous throughout the year
- Energy is available from nature at no cost

Disadvantages

- Thermal efficiency is low due to small temperature gradient
- Capital cost is high due to necessity of H.E., boiler and condenser

Tidal energy

- Tide → periodic rise and fall of sea water caused by interaction of gravitational attraction of sun and moon

- Tidal energy → energy comes from tide in form of potential & K.E.

- Highest level of tidal water → high tide or flood

  Lowest level of tidal water → low tide or ebb

- During high tide (rise of tide) → tidal basin is filled up

  During low tide (fall of tide) → tidal basin is emptied

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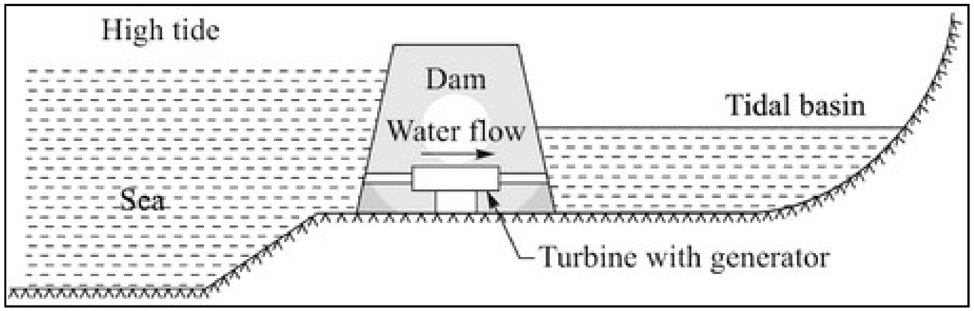
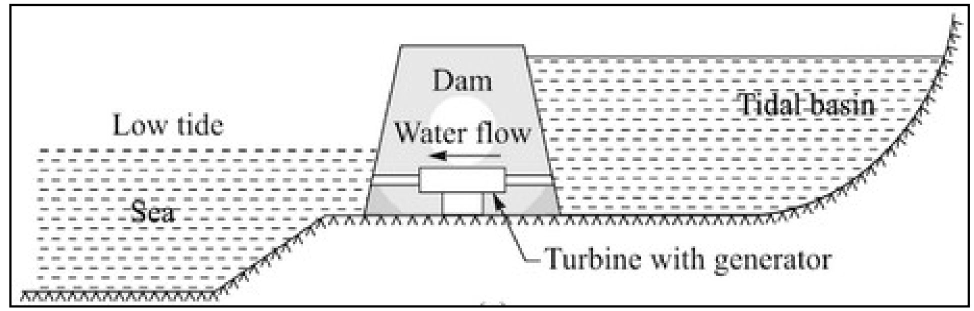
Single pool tidal energy conversion





- High tide and water flows in from sea to tidal basin
- While flowing it runs the turbine and generate electricity

- Low tide and water flows out from tidal basin to sea
- While flowing it runs the turbine and generate electricity

Contd….

Advantages of Tidal energy

- Free from pollution
- Best suited to meet peak power demands
- Superior to hydel energy as it does not depend on rains

Disadvantages

- Costly
- Power generation is not continuous and depends on the capacity of tidal basin